Foundations of K-means clustering

[K-means clustering]

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

1.0 Content Block

k-means implicitly assumes that the ordering of the input data set does not matter. The bilateral filter is similar to k-means and mean shift in that it maintains a set of data points that are iteratively replaced by means. However, the bilateral filter restricts the calculation of the (kernel weighted) mean to include only points that are close in the ordering of the input data. This makes it applicable to problems such as image denoising, where the spatial arrangement of pixels in an image is of critical importance.

2.0 Content Block

The slow "standard algorithm" for k-means clustering, and its associated expectation–maximization algorithm, is a special case of a Gaussian mixture model, specifically, the limiting case when fixing all covariances to be diagonal, equal and have infinitesimal small variance. Instead of small variances, a hard cluster assignment can also be used to show another equivalence of k-means clustering to a special case of "hard" Gaussian mixture modelling. This does not mean that it is efficient to use Gaussian mixture modelling to compute k-means, but just that there is a theoretical relationship, and that Gaussian mixture modelling can be interpreted as a generalization of k-means; on the contrary, it has been suggested to use k-means clustering to find starting points for Gaussian mixture modelling on difficult data.

3.0 Content Block

a r g m i n S ■ ∑ i = 1 k ∑ x ∈ S i ■ x − μ i ■ 2 . {\displaystyle \mathop {\operatorname {arg\,min} } _{\mathbf {S} }\sum _{i=1}^{k}\sum _{\mathbf {x} \in S_{i}}\left\|\mathbf {x} -{\boldsymbol {\mu }}_{i}\right\|^{2}.}

4.0 Content Block

In biological applications, the limitations of standard k-means clustering are often addressed by adapting the distance measures used and objective functions to better reflect the structure of experimental data. Rather than relying on Euclidean distance and variance-based measures of cluster scatter, alternative similarity metrics such as the Jaccard index are sometimes used when analyzing genomic datasets with k-means. In these contexts, clustering may be performed directly on distance matrices, and medoids (representative data points that have the smallest average distance to every other point within a cluster) may be selected as cluster centers instead of arithmetic means. This allows clustering methods to capture patterns of shared genomic features or co-occurrence that are not well represented in a traditional Euclidean analysis. Clustering is widely used in biological data analysis to identify patterns in high-dimensional datasets such as gene expression profiles and genomic interaction matrices. Techniques such as k-means clustering partition observations into groups based on similarity, allowing researchers to detect structure in complex datasets. In gene expression studies, clustering is commonly used to group genes with similar expression profiles, often revealing co-regulated genes and underlying biological interactions. In addition to gene expression analysis, clustering approaches are applied to chromatin interaction data to identify regions of coordinated activity. Similarity measures such as the Jaccard index can be used to quantify shared features between observations, and clustering of the resulting distance matrices can reveal structural organization within the genome. These patterns are commonly visualized using heatmaps, where clustered regions correspond to domains of interaction or co-segregation, providing insight into the regulatory organization of the genome. In such analyses, clustering of similarity or co-detection matrices can reveal groups of genomic windows that exhibit coordinated behavior across experimental conditions. Clustering methods provide a practical method for uncovering biologically meaningful structure from complex datasets. By grouping observations with similar patterns of genomic detection, expression, or interaction, these techniques can reveal functional relationships, spatial organization within the nucleus, and regulatory behavior. In many applications, the resulting clusters correspond to distinct biological states or structural domains, further illustrating the broad applicability of k-means clustering.

5.0 Content Block

Variations Jenks natural breaks optimization: k-means applied to univariate data k-medians clustering uses the median in each dimension instead of the mean, and this way minimizes L 1 {\displaystyle L_{1}} norm (Taxicab geometry). k-medoids (also: Partitioning Around Medoids, PAM) uses the medoid instead of the mean, and this way minimizes the sum of distances for arbitrary distance functions. Fuzzy C-Means Clustering is a soft version of k-means, where each data point has a fuzzy degree of belonging to each cluster. Gaussian mixture models trained with expectation–maximization algorithm (EM algorithm) maintains probabilistic assignments to clusters, instead of deterministic assignments, and multivariate Gaussian distributions instead of means. k-means++ chooses initial centers in a way that gives a provable upper bound on the WCSS objective. The filtering algorithm uses k-d trees to speed up each k-means step. Some methods attempt to speed up each k-means step using the triangle inequality. Escape local optima by swapping points between clusters. The Spherical k-means clustering algorithm is suitable for textual data. Hierarchical variants such as Bisecting k-means, X-means clustering and G-means clustering repeatedly split clusters to build a hierarchy, and can also try to automatically determine the optimal number of clusters in a dataset. Internal cluster evaluation measures such as cluster silhouette can be helpful at determining the number of clusters. Minkowski weighted k-means automatically calculates cluster specific feature weights, supporting the intuitive idea that a feature may have different degrees of relevance at different features. These weights can also be used to re-scale a given data set, increasing the likelihood of a cluster validity index to be optimized at the expected number of clusters. Mini-batch k-means: k-means variation using "mini batch" samples for data sets that do not fit into memory. Otsu's method

6.0 Content Block

Applications k-means clustering is rather easy to apply to even large data sets, particularly when using heuristics such as Lloyd's algorithm. It has been successfully used in market segmentation, computer vision, and astronomy among many other domains. It often is used as a preprocessing step for other algorithms, for example to find a starting configuration.

7.0 Content Block

Software implementations Different implementations of the algorithm exhibit performance differences, with the fastest on a test data set finishing in 10 seconds, the slowest taking 25,988 seconds (~7 hours). The differences can be attributed to implementation quality, language and compiler differences, different termination criteria and precision levels, and the use of indexes for acceleration.

8.0 Content Block

Δ (x , n , m) = ■ S n ■ ■ S n ■ – 1 · ■ μ n – x ■ 2 – ■ S m ■ ■ S m ■ + 1 · ■ μ m – x ■ 2 . {\displaystyle \Delta (x,n,m)={\frac {\mid S_{n}\mid }{\mid S_{n}\mid -1}}\cdot \lVert \mu _{n}-x\rVert ^{2}-{\frac {\mid S_{m}\mid }{\mid S_{m}\mid +1}}\cdot \lVert \mu _{m}-x\rVert ^{2}.}